

Diagram and Data Dictionary

Internal database structure of the Data Management Portal

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Tables: 18

Columns: 195

1. How to read this document

Every table is created by `migrate()`, in `backend/db.php`, which runs on each request and is idempotent. The real name carries the configured prefix -- with the default `gdt`, `bancos` becomes `gdt_bancos`.

There are no FOREIGN KEYS -- integrity between related tables is enforced in application code, keeping the schema portable across MySQL, PostgreSQL, SQL Server and SQLite. The `bancos` table has a UNIQUE index on `(servidor, nome)`: several databases per server, but no two with the same name on the same host.

2. Overview

Accounts, session and security

Table	Cols	Description
<code>usuarios</code>	15	Portal accounts, with role and preferences.
<code>usuarios_foto</code>	3	Profile photo, in a separate table.
<code>tentativas_login</code>	7	Failed login counters.
<code>tokens_revogados</code>	2	Tokens invalidated before expiry.
<code>mfa_codigos</code>	9	Two-step verification codes (HMAC only).
<code>auditoria_log</code>	9	Audit trail: who, what, when and from which IP.

Governance modules

Table	Cols	Description
<code>bancos</code>	14	Central registry of databases, reused across the other modules instead of retyping server and name.

Table	Cols	Description
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Portal configuration

Table	Cols	Description
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config_projeto	5	Portal identity: name, logo and session.

3. Dictionary

usuarios

Portal accounts, with role and preferences.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
username	VARCHAR(50) NOT NULL UNIQUE	Login name, unique in the portal.
nome_completo	VARCHAR(120)	Name shown in the interface.
password_hash	VARCHAR(255) NOT NULL	Password as bcrypt. Never leaves via the API.
role	VARCHAR(20) NOT NULL DEFAULT 'leitura'	Role: leitura, escrita, admin or master.
criado_em	date and time	Creation date and time.
atualizado_em	date and time	Date and time of the last change.
modulos_permitidos	long text (TEXT / NVARCHAR(MAX))	Allowed modules, comma separated.
email	VARCHAR(150)	E-mail, used for the verification code.
mfa_ativo	INT NOT NULL DEFAULT 0	1 when two-step verification is on.
cor_tema	VARCHAR(20) NOT NULL DEFAULT 'padrao'	Color palette chosen by the user.
estilo_side	VARCHAR(20) NOT NULL DEFAULT 'claro'	Sidebar appearance.
ativo	INT NOT NULL DEFAULT 1	0 disables the account without erasing history.
must_change_password	INT NOT NULL DEFAULT 0	1 forces a password change on next sign-in.
tokens_valid_after	date and time	Mass revocation cutoff.

usuarios_foto

Profile photo, in a separate table.

Column	Type	Description
usuario_id	INT NOT NULL PRIMARY KEY	User who owns the photo (primary key, no auto-increment).
foto	very long text (MEDIUMTEXT on MySQL)	Image as a data URI (base64), 256x256.
atualizado_em	date and time	Date and time of the last change.

tentativas_login

Failed login counters.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
usuario	VARCHAR(150) NOT NULL	Login related to the record.
ip	VARCHAR(45) NOT NULL	Source IP address.
tentativas	INT NOT NULL DEFAULT 0	Verification attempts (max 5).
primeira_tentativa	date and time	Module field.
ultima_tentativa	date and time	Module field.
bloqueado_ate	date and time	Module field.

tokens_revogados

Tokens invalidated before expiry.

Column	Type	Description
jti	VARCHAR(64) NOT NULL PRIMARY KEY	Identifier of the revoked token.
expira_em	date and time	Code validity (10 minutes).

mfa_codigos

Two-step verification codes (HMAC only).

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
usuario	VARCHAR(150) NOT NULL	Login related to the record.
token	VARCHAR(64) NOT NULL	Verification session token.
codigo_hash	VARCHAR(64) NOT NULL	HMAC of "token code".
tentativas	INT NOT NULL DEFAULT 0	Verification attempts (max 5).
expira_em	date and time	Code validity (10 minutes).
criado_em	date and time	Creation date and time.
reenvios	INT NOT NULL DEFAULT 0	Module field.
ultimo_reenvio	DATETIME	Module field.

auditoria_log

Audit trail: who, what, when and from which IP.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
tabela	VARCHAR(60) NOT NULL	Related table.
registro_id	INT	Module field.
acao	VARCHAR(40) NOT NULL	Module field.
usuario	VARCHAR(150)	Login related to the record.
dados_antes	long text (TEXT / NVARCHAR(MAX))	Module field.
dados_depois	long text (TEXT / NVARCHAR(MAX))	Module field.
ip	VARCHAR(45)	Source IP address.
criado_em	date and time	Creation date and time.

bancos

Central registry of databases, reused across the other modules instead of retyping server and name.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
nome	VARCHAR(150) NOT NULL	Record name.
servidor	VARCHAR(150)	Server hostname (without the database).
motor	VARCHAR(40)	Database engine (MySQL, PostgreSQL, SQL Server...).
ambiente	VARCHAR(40)	Environment: Production, Staging, Development, Test.
host	VARCHAR(200)	IPv4 address of the database.
porta	VARCHAR(10)	Port of the IPv4 address.
ipv6	VARCHAR(200)	IPv6 address of the database.
porta_ipv6	VARCHAR(10)	Port of the IPv6 address.
responsavel	VARCHAR(150)	Person or team responsible.
obs	long text (TEXT / NVARCHAR(MAX))	Additional notes.
criado_por	VARCHAR(150)	Login of whoever created the record (set by the server).
criado_em	date and time	Creation date and time.
atualizado_em	date and time	Date and time of the last change.

acessos

Inventory of access granted to databases and systems.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
data	DATE	Module field.
usuario	VARCHAR(150)	Login related to the record.
tipo	VARCHAR(50)	Classification (from Cadastro or fixed list).
objeto	VARCHAR(200)	Database selected from the Databases registry.
nivel	VARCHAR(100)	Module field.
justificativa	long text (TEXT / NVARCHAR(MAX))	Module field.
solicitante	VARCHAR(150)	Module field.
aprovador	VARCHAR(150)	Module field.
revisao	DATE	Module field.
status	VARCHAR(30) DEFAULT 'Ativo'	Current status (from Cadastro).
obs	long text (TEXT / NVARCHAR(MAX))	Additional notes.
criado_em	date and time	Creation date and time.
atualizado_em	date and time	Date and time of the last change.
servidor	VARCHAR(150)	Server hostname (without the database).
natureza	VARCHAR(20)	Nature of the access account: User or Service.
criado_por	VARCHAR(150)	Login of whoever created the record (set by the server).

mudancas

Log of changes to data structures.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
codigo	VARCHAR(30)	Module field.
data	DATE	Module field.
ambiente	VARCHAR(30)	Environment: Production, Staging, Development, Test.
tipo	VARCHAR(50)	Classification (from Cadastro or fixed list).
descricao	long text (TEXT / NVARCHAR(MAX))	Free description of the record.
script	VARCHAR(200)	Module field.
rollback	long text (TEXT / NVARCHAR(MAX))	Module field.
solicitante	VARCHAR(150)	Module field.
aprovador	VARCHAR(150)	Module field.
status	VARCHAR(30)	Current status (from Cadastro).
resultado	long text (TEXT / NVARCHAR(MAX))	Module field.
criado_em	date and time	Creation date and time.
atualizado_em	date and time	Date and time of the last change.
criado_por	VARCHAR(150)	Login of whoever created the record (set by the server).
bancos	long text (TEXT / NVARCHAR(MAX))	Databases affected by the change (multiple selection, comma separated).

mudancas_objetos

Objects affected by each change.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
mudanca_id	INT NOT NULL	Module field.
nome	VARCHAR(200) NOT NULL	Record name.
tipo	VARCHAR(50)	Classification (from Cadastro or fixed list).
criado_em	date and time	Creation date and time.

backup_politicas

Backup policy per database.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
banco	VARCHAR(150)	Related database (from the Databases registry).
criticidade	VARCHAR(30)	Criticality level (from Cadastro).
tipo	VARCHAR(50)	Classification (from Cadastro or fixed list).
frequencia	VARCHAR(100)	Routine periodicity.
horario	VARCHAR(50)	Frequency interval -- how often it repeats.
retencao	VARCHAR(100)	Module field.
local	VARCHAR(250)	Module field.
rpo	VARCHAR(50)	Module field.
rto	VARCHAR(50)	Module field.
responsavel	VARCHAR(150)	Person or team responsible.
criado_em	date and time	Creation date and time.
atualizado_em	date and time	Date and time of the last change.
criado_por	VARCHAR(150)	Login of whoever created the record (set by the server).

restore_testes

Restore tests performed.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
data	DATE	Module field.
banco	VARCHAR(150)	Related database (from the Databases registry).
backup	VARCHAR(150)	Module field.
tempo	VARCHAR(50)	Module field.
resultado	VARCHAR(30)	Module field.
por	VARCHAR(150)	Module field.
obs	long text (TEXT / NVARCHAR(MAX))	Additional notes.
criado_em	date and time	Creation date and time.
atualizado_em	date and time	Date and time of the last change.
criado_por	VARCHAR(150)	Login of whoever created the record (set by the server).

dicionario_dados

Business dictionary of the managed databases columns.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
banco	VARCHAR(150)	Related database (from the Databases registry).
schema_nome	VARCHAR(150)	Table schema.
tabela	VARCHAR(150)	Related table.
coluna	VARCHAR(150)	Column described.
tipo_dado	VARCHAR(100)	Module field.
permite_nulo	VARCHAR(10)	Module field.
descricao	long text (TEXT / NVARCHAR(MAX))	Free description of the record.
classificacao	VARCHAR(50)	Module field.
origem	VARCHAR(150)	Module field.
obs	long text (TEXT / NVARCHAR(MAX))	Additional notes.
criado_em	date and time	Creation date and time.
atualizado_em	date and time	Date and time of the last change.
servidor	VARCHAR(150)	Server hostname (without the database).
criado_por	VARCHAR(150)	Login of whoever created the record (set by the server).

integracoes

Integrations between systems.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
nome	VARCHAR(200)	Record name.
origem	VARCHAR(150)	Module field.
ip_origem	VARCHAR(45)	Module field.
destino	VARCHAR(150)	Module field.
ip_destino	VARCHAR(45)	Module field.
tipo	VARCHAR(60)	Classification (from Cadastro or fixed list).
direcao	VARCHAR(30)	Module field.
mecanismo	VARCHAR(150)	Module field.
frequencia	VARCHAR(30)	Routine periodicity.
dados_trafegados	long text (TEXT / NVARCHAR(MAX))	Module field.
classificacao	VARCHAR(50)	Module field.
criticidade	VARCHAR(30)	Criticality level (from Cadastro).
resp_tecnico	VARCHAR(150)	Module field.
resp_negocio	VARCHAR(150)	Module field.
ambiente	VARCHAR(30)	Environment: Production, Staging, Development, Test.
status	VARCHAR(30)	Current status (from Cadastro).
ultima_revisao	DATE	Module field.
obs	long text (TEXT / NVARCHAR(MAX))	Additional notes.
criado_em	date and time	Creation date and time.
atualizado_em	date and time	Date and time of the last change.
criado_por	VARCHAR(150)	Login of whoever created the record (set by the server).

jobs

Scheduled routines: command, steps, schedules.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
nome	VARCHAR(200)	Record name.
tipo	VARCHAR(60)	Classification (from Cadastro or fixed list).
servidor	VARCHAR(150)	Server hostname (without the database).
banco	VARCHAR(150)	Related database (from the Databases registry).
descricao	long text (TEXT / NVARCHAR(MAX))	Free description of the record.
comando	long text (TEXT / NVARCHAR(MAX))	Job steps as JSON.
frequencia	VARCHAR(60)	Routine periodicity.
horario	VARCHAR(60)	Frequency interval -- how often it repeats.
agendamentos	long text (TEXT / NVARCHAR(MAX))	List of windows (start and end) as JSON.
criticidade	VARCHAR(30)	Criticality level (from Cadastro).
status	VARCHAR(30)	Current status (from Cadastro).
responsavel	VARCHAR(150)	Person or team responsible.
obs	long text (TEXT / NVARCHAR(MAX))	Additional notes.
criado_por	VARCHAR(150)	Login of whoever created the record (set by the server).
criado_em	date and time	Creation date and time.
atualizado_em	date and time	Date and time of the last change.

config_tipos

Editable lists used in the forms.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
categoria	VARCHAR(40) NOT NULL	Module field.
nome	VARCHAR(120) NOT NULL	Record name.
criado_em	date and time	Creation date and time.

config_email

SMTP settings; the password uses AES-256-GCM.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
host	VARCHAR(150)	IPv4 address of the database.
porta	INT	Port of the IPv4 address.
seguranca	VARCHAR(10) DEFAULT 'tls'	Module field.
usuario	VARCHAR(150)	Login related to the record.
senha_cifrada	long text (TEXT / NVARCHAR(MAX))	Module field.
remetente_nome	VARCHAR(120)	Module field.
remetente_email	VARCHAR(150)	Module field.
testado_ok	INT NOT NULL DEFAULT 0	1 after a successful test send.
atualizado_em	date and time	Date and time of the last change.

config_projeto

Portal identity: name, logo and session.

Column	Type	Description
id	integer, primary key, auto-increment	Unique record identifier.
nome_projeto	VARCHAR(160)	Module field.
logo_data	very long text (MEDIUMTEXT on MySQL)	Portal logo as a data URI. MEDIUMTEXT on MySQL.
atualizado_em	date and time	Date and time of the last change.
timeout_inatividade_min	INT NOT NULL DEFAULT 30	Minutes of inactivity before the session ends.

4. Relationships

From	To	Rule
acessos.objeto / servidor	bancos.nome / servidor	Bank picked from the registry fills both.
mudancas.bancos	bancos.nome	Databases affected, comma separated.
backup / restore / dicionario / jobs .banco	bancos.nome	Database selected from the registry.
usuarios_foto.usuario_id	usuarios.id	One photo per user; removed with the user.
mudancas_objetos.mudanca_id	mudancas.id	Objects affected; removed with the parent.
auditoria_log.usuario	usuarios.username	By login, so the trail survives deletion.